



Tuka Curious Minds

MASTER ROBO WARS DESIGN

CONCEPT SUBMISSION

Robot Name: TURBO Ceros-T

Category: Robo Wars

Weight Classification: 5.kg Max Limit

Footprint Matrix: 420mm (Width) x 420mm (Length) x 55mm (Sleek Casing Height)

Inspiration Concepts discussed, sketched and AI generated to get to try generate a CAD design.

*Note the team are new to CAD, was a steep learning curve but tried to navigate through AI assistance, senior TCM members and YouTube tutorials

Version #1

Scrapped not feasible



We were thinking of doing a race car initially and reconsidered the material and components we had. We then thought there are 3 females and 1 male on the team with male coaches also and changed our mind to try out Robo Wars as females are under represented in this male dominated competition and career right?

Let's do this! Let's GO!

Game Play



Max damage to opponents??

Defence

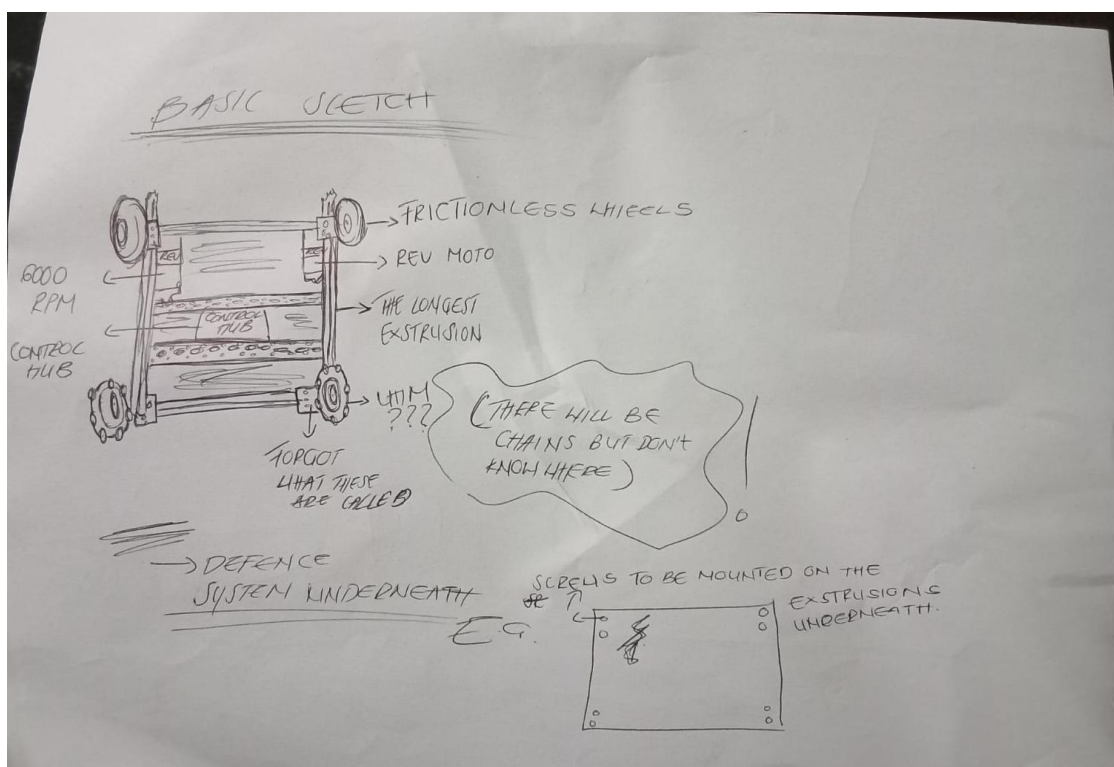
Speed – [running away!]

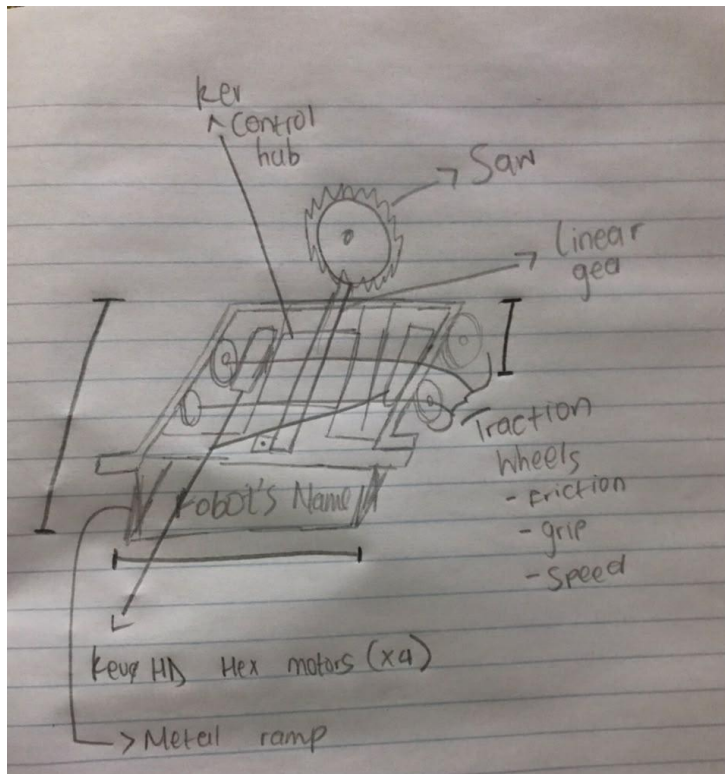
Last one standing...

Version 2

To consider – component parts such as traction wheels, extrusions vs c-channel, available motors.

Weapon system – 2 – 3 blades mounted on front of robot???





Version 3

Material - GoBuilder /Rev

Components. C channels that will hold onto the four traction wheels, because of their grip, speed and friction and then two on each side.

Motors - Single motor that drives both the wheels on each side.

- REV HTX motor, 6000 RPM, or

- GoBuilder 1:1 Yellow Jacket

Chain – Join that using a chain

Control - REV control hub

Weapons - solenoid that will stick out of the robot using a linear gear.

Front - A metal ramp that can pull, push, or lift up the opponent's robot and flip it over. motor, 6000 RPM, and we must make sure each motor has a good torque + speed for wheels

FINAL DESIGN THOUGHTS.....

Damage

Speed

FINAL CHOICE DESIGN

So we took all discussions and sketches and ran through AI to get a feel of what it could look like and compared all possible designs.



LET'S DESIGN TURBO CEROS-T!



Now for Blender/CAD attempt

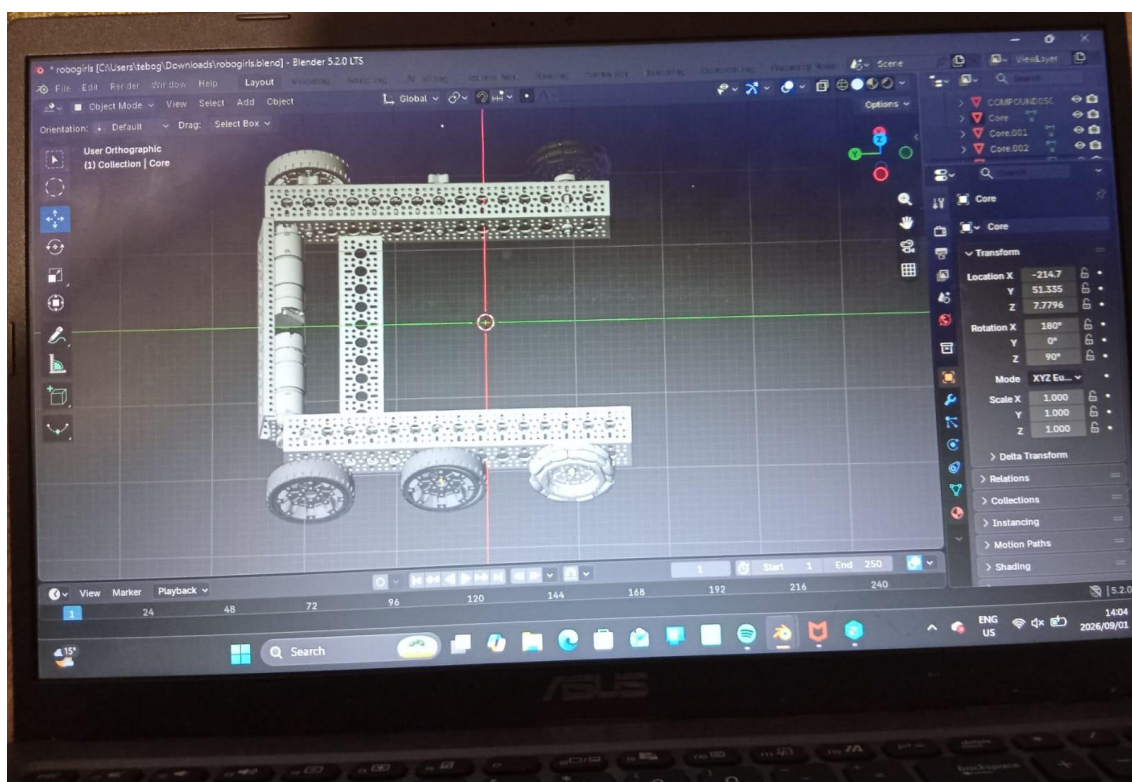
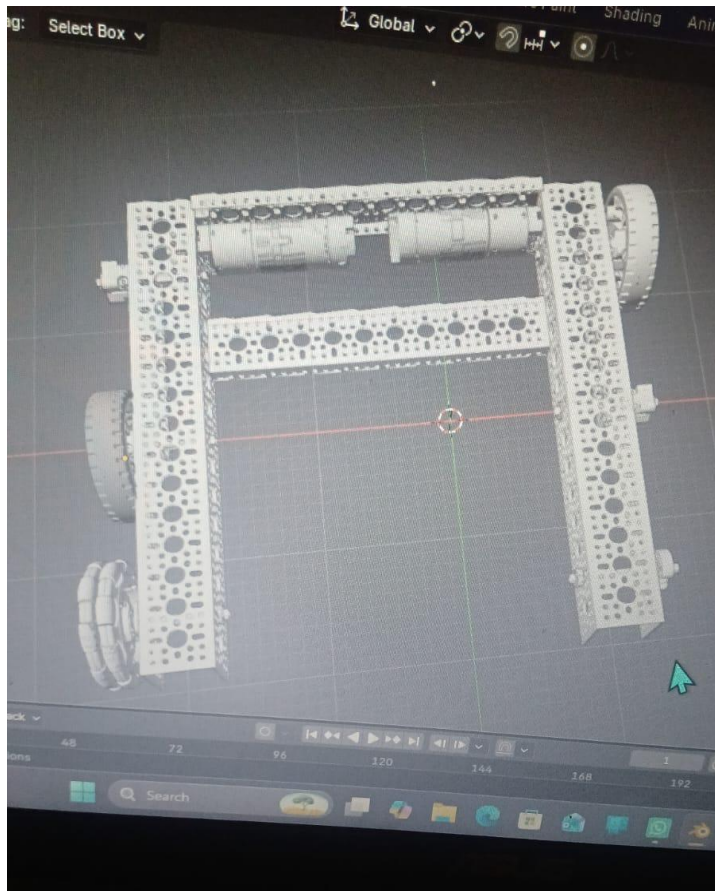
CAD #1



Thank you to senior coach Bonolo for getting us started and coaching session, but this is not easy as he makes it look!

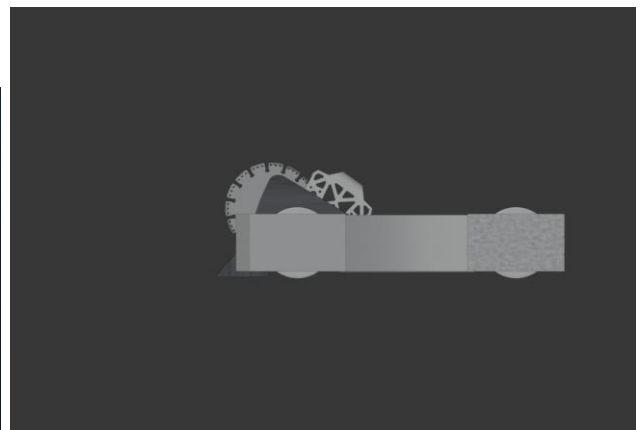
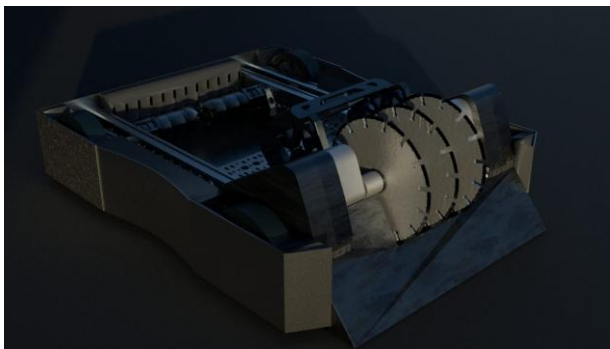
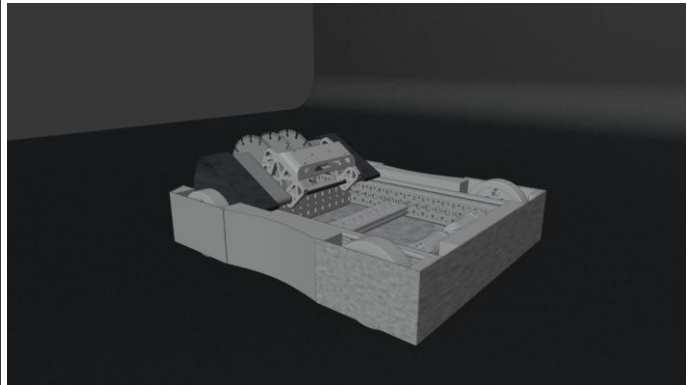
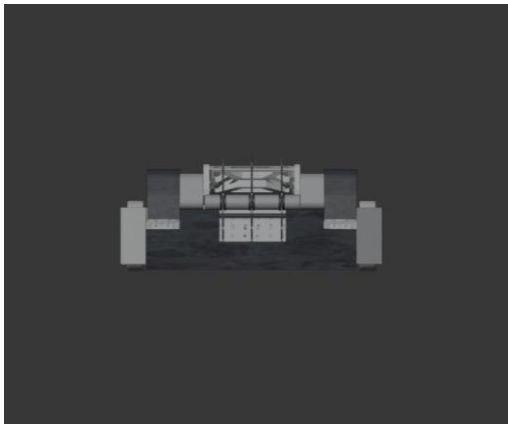
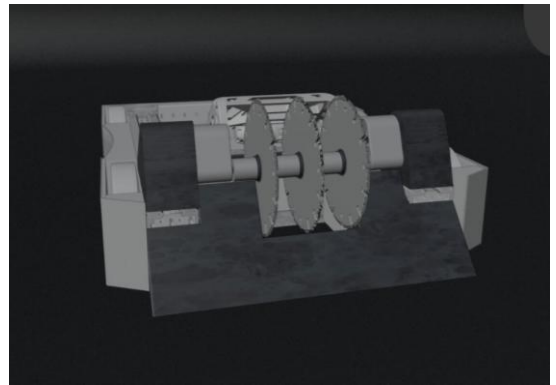
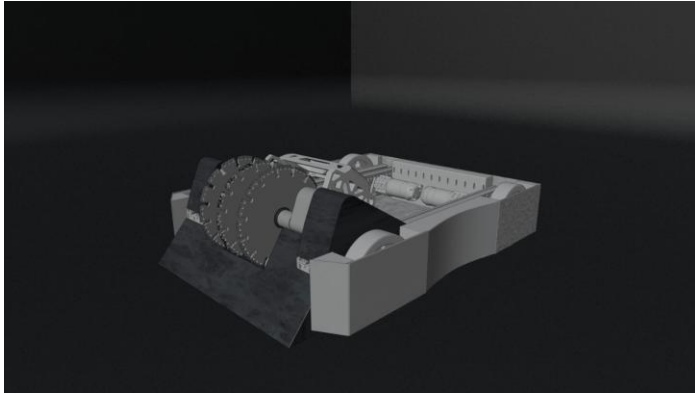
We tried our best....

CAD #2



Not sure if the problem was the laptop, but the ASUS we using keeps stalling, crashing, restarting. This is how far we got

CAD #3



Thank you to Coach Bonolo for stepping in and assisting with final render version!

Should we get selected for next round, we hope to get a chance and opportunity to learn the basics and fundamentals of CAD as this was our first attempt as the girls squad trying to understand how to do this.

Thank you for considering TURBO Ceros-T to become one of the bots in the arena,

Making it through to the finals gives us a chance to compete against the boys!

Our team aspirations are to learn more of and gain exposure to programming using different languages, mechatronics and mechanics.

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